DataLineageML: Causal Data Provenance for AI Safety

A Research Framing Document

Oluwafemi Philip Adejumobi · DeepFly Tech Labs · March 2026

Abstract. Modern AI systems fail silently. A crop yield model allocates subsidy to the wrong farmers. A loan model denies credit along racial lines. A clinical model underperforms for women. In each case, the failure is measurable, but the cause is invisible. Existing observability tools (MLflow, Weights & Biases, LangSmith) record what happened inside an AI pipeline. None of them answer the question that actually matters: which specific data transformation caused this outcome, and how do we prove it? DataLineageML addresses this gap by combining automated data provenance tracking with causal attribution, tracing safety and fairness failures back to the exact pipeline step responsible, and verifying the fix through counterfactual replay.

1. THE PROBLEM: INVISIBLE CAUSALITY IN AI PIPELINES

Every AI system that affects real people is built on a sequence of data transformations. Raw sensor readings are cleaned. Missing values are dropped. Features are engineered. Datasets are merged. Each of these steps is a decision, and each decision can introduce, amplify, or mask bias. By the time a trained model is evaluated and found to be unfair, the transformation responsible may be buried three steps back in a pipeline that nobody remembers clearly.

This is not a theoretical problem. In Nigeria's agricultural sector, where government subsidies, extension officer resources, and market access are increasingly allocated by data-driven systems, a biased ML model does not merely produce a wrong number. It decides which smallholder farmer in Oyo State receives fertiliser support and which does not. The consequences are material, multi-year, and largely irreversible.

The tools AI teams currently use to address this are inadequate. Observability platforms record pipeline execution telemetry, what ran, how long it took, what the inputs and outputs looked like. Fairness toolkits measure bias in model outputs. But neither connects the two. The gap between "the model is biased" and "this specific data step caused the bias" remains unbridged. As a result, when a fairness audit finds a problem, the remediation process is manual, expensive, and often wrong, teams fix the wrong thing.

> "The challenge is not measuring bias in a model output. The challenge is tracing that bias back through the pipeline to the decision that caused it, and proving the fix works before the model reaches production."

This is the problem DataLineageML is designed to solve. Not observability. Not evaluation. Causal attribution: the automated identification of the pipeline step responsible for a measurable safety or fairness failure, with counterfactual evidence to verify the remedy.

2. THE APPROACH: PROVENANCE GRAPHS WITH CAUSAL ATTRIBUTION

DataLineageML instruments ML pipelines at the function level using a lightweight Python decorator. Every transformation step, data loading, cleaning, feature engineering, model training, is automatically logged with cryptographic hashes of its inputs and outputs, timing metadata, and a statistical snapshot of the data at that point in the pipeline.

Three capabilities distinguish this approach from existing tools:

- **Statistical provenance.** Snapshot-level demographic tracking. At every pipeline step, DataLineageML records the statistical distribution of sensitive attributes, gender, age group, geographic zone, farm size, not just the data hash. This produces a longitudinal demographic profile of the dataset as it moves through the pipeline.
- **Causal attribution.** Rather than simply logging what happened, DataLineageML computes distribution shift scores between adjacent pipeline steps using Jensen-Shannon divergence for categorical attributes and the Kolmogorov-Smirnov test for continuous ones. Steps with high shift scores on sensitive attributes are flagged as candidate causal steps.

- **Counterfactual verification.** When a candidate step is identified, DataLineageML re-executes the pipeline from that step with a modified transformation, re-evaluates the fairness metric, and reports the delta. A finding is only reported when counterfactual evidence confirms the attribution.

The system is built on a zero-dependency SQLite core, operates entirely offline without cloud connectivity, and is designed for deployment in resource-constrained environments, including government agricultural data systems in sub-Saharan Africa where cloud data transfer is legally restricted and network reliability is limited.

3. A CONCRETE EXAMPLE: OYO STATE AGRICULTURAL SUBSIDY ALLOCATION

3.1 Background

In 2025, DeepFly Tech Labs secured a ₦2.45 million grant from the Oyo State Ministry of Agriculture to design and deploy a data collection system enabling agricultural extension officers to systematically collect farm-level data across all 33 local government areas, forming the first centralised, state-wide database of crop, livestock, and agro-allied operations in Oyo State.

A downstream use of this data is a crop yield forecasting model that informs subsidy allocation: which farmers receive fertiliser, irrigation support, and market linkage assistance. The pipeline ingests farm registration records, field survey data, satellite imagery, and historical yield records. A bias failure in this pipeline means female smallholder farmers, who operate 40% of farms in Oyo State but are systematically under-represented in formal registration systems, receive less support than their farming performance warrants.

3.2 The Failure

An audit of the subsidy allocation model found that female-headed farm households were 34% less likely to be recommended for fertiliser support than male-headed households with equivalent yield histories. The model's overall accuracy was strong. The bias was invisible in aggregate metrics. The team did not know where in the six-step preprocessing pipeline the disparity originated.

3.3 What DataLineageML Found

With DataLineageML instrumented across the pipeline, each step logged demographic snapshots of the farm registration dataset. The distribution shift detector identified the following at the clean_data step:

```
Step: clean_data (df.dropna())
  Before:  female-headed farms = 38.4%  (n = 12,847)
  After:   female-headed farms = 22.1%  (n =  7,403)
  Jensen-Shannon divergence on gender: 0.81  [HIGH]
  Attributed cause: dropna() removes rows with missing 'land_title'
  Context: formal land titles held by 11% of female vs 67% of male farmers
```

The attribution finding was precise: a single `df.dropna()` call, standard, unremarkable, the default first step in nearly every data science notebook, was silently removing 42% of female-headed farm records because women in Oyo State are far less likely to hold formal land title documentation, which was a required field in the registration form.

The counterfactual replayer re-ran the pipeline replacing `dropna()` with a stratified imputation strategy that preserved demographic representation, and re-evaluated the bias metric:

```
Counterfactual result:
  Original gender bias score:     0.34
  Post-fix gender bias score:     0.09
  Bias reduction:                 74%
  Subsidy allocation equity gap:  34% → 6%
  Model accuracy change:          -0.3%  [negligible]
```

The fix took one afternoon. The discovery, without DataLineageML, had been eluding the team for three months of manual audit.

4. SIGNIFICANCE AND SCOPE

The Oyo State example is not unusual. It is representative of a broad class of AI safety failures: ones that are caused by a routine engineering decision applied to data with structural inequalities that the engineer did not know were there. Formal land title gaps, gender-differentiated mobile phone ownership, age-correlated literacy rates in survey responses, these are properties of the society, not of the algorithm, and they do not appear in accuracy metrics.

DataLineageML's contribution is operational: it makes the invisible visible, makes attribution automatic, and makes the verification of remediation rigorous. The research contribution is formal: establishing the theoretical connection between pipeline-level data provenance graphs and causal inference frameworks for bias attribution, a connection that has not previously been formalised in the AI safety literature.

The scope extends beyond agricultural AI. The same pipeline architecture, track, snapshot, attribute, replay, applies to:

- Fine-tuned language models, where training data imbalances produce demographic disparities in text generation.
- RAG systems, where document corpus composition determines whose knowledge is retrievable and whose is not.
- Multi-agent systems, where data provenance must cross agent boundaries to trace failures across distributed pipelines.

In each of these settings, the question is the same: which specific data decision caused this safety failure? DataLineageML is the first tool designed to answer it automatically, verifiably, and without requiring a cloud account or an ML infrastructure team.